

## ITHACA 13 E, NY, USA

WMO: 725270

Lat: 42.4401N

Lon: 76.2462W

Elev: 1228

StdP: 14.06

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 64758

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |            | Humidification DP/MCDB and HR |            |             |             |            |            | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|------------|-------------------------------|------------|-------------|-------------|------------|------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |            | 99.6%                         |            |             | 99%         |            |            | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%        | DP                            | HR         | MCDB        | DP          | HR         | MCDB       | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-4.1 | (c)<br>1.1 | (d)<br>-11.1                  | (e)<br>3.2 | (f)<br>-4.0 | (g)<br>-6.5 | (h)<br>4.1 | (i)<br>1.2 | (j)<br>18.6           | (k)<br>22.5 | (l)<br>17.1 | (m)<br>21.3 | (n)<br>3.6            | (o)<br>N/A | (p)<br>0.495 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.2                   | 85.7            | 70.6 | 83.0 | 69.3 | 80.4 | 68.3 | 73.3                | 82.1 | 71.8 | 79.5 | 70.4 | 77.6 | 5.7                  | N/A  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 70.5                            | 117.4 | 78.1 | 69.2 | 112.2 | 76.3 | 67.9 | 107.4 | 75.0 | 37.8          | 82.5 | 36.3 | 79.8 | 35.1 | 77.1 | 77.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 16.7              | 14.8 | 13.1 | -12.4                      | 90.1 | 6.3                | 2.9 | -16.9                                              | 92.2 | -20.6      | 93.9 | -24.2      | 95.5 | -28.8      | 97.7 |  |  |
|                   |      |      | -13.9                      | 75.9 | 5.6                | 1.4 | -18.0                                              | 76.9 | -21.3      | 77.7 | -24.5      | 78.5 | -28.5      | 79.4 |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
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|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
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|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
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|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
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|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
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|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 45.9   | 22.4  | 22.9  | 31.3  | 44.1  | 55.4  | 63.6  | 67.9  | 65.8  | 59.7  | 49.2  | 37.8  | 28.6  |
|                                                                        | DBStd       | 18.45  | 12.18 | 10.92 | 11.05 | 9.09  | 8.66  | 6.39  | 5.39  | 5.54  | 7.33  | 8.42  | 9.24  | 9.58  |
|                                                                        | HDD50       | 3636   | 856   | 759   | 589   | 220   | 43    | 1     | 0     | 0     | 6     | 121   | 377   | 664   |
|                                                                        | HDD65       | 7314   | 1321  | 1179  | 1046  | 627   | 318   | 102   | 33    | 59    | 192   | 493   | 816   | 1128  |
|                                                                        | CDD50       | 2121   | 0     | 0     | 8     | 44    | 210   | 408   | 556   | 489   | 298   | 96    | 11    | 1     |
|                                                                        | CDD65       | 325    | 0     | 0     | 0     | 1     | 20    | 60    | 123   | 83    | 34    | 4     | 0     | 0     |
|                                                                        | CDH74       | 2860   | 0     | 0     | 0     | 39    | 247   | 550   | 1084  | 637   | 283   | 20    | 0     | 0     |
|                                                                        | CDH80       | 625    | 0     | 0     | 0     | 8     | 44    | 128   | 270   | 120   | 54    | 1     | 0     | 0     |
|                                                                        | WSAvg       | 5.8    | 7.0   | 7.3   | 7.3   | 7.0   | 5.4   | 4.6   | 3.9   | 4.0   | 4.2   | 5.3   | 6.3   | 7.1   |
|                                                                        | PrecAvg     | 40.5   | 2.7   | 2.2   | 3.2   | 3.3   | 3.3   | 4.3   | 4.0   | 4.0   | 3.9   | 3.9   | 2.8   | 3.0   |
| (15) Precipitation                                                     | PrecMax     | 49.1   | 4.8   | 4.6   | 6.1   | 8.0   | 6.7   | 9.1   | 8.8   | 7.8   | 6.9   | 7.8   | 4.3   | 5.5   |
|                                                                        | PrecMin     | 31.1   | 1.2   | 1.1   | 1.2   | 0.8   | 1.2   | 1.4   | 1.5   | 1.9   | 2.2   | 1.2   | 1.0   | 1.4   |
|                                                                        | PrecStd     | 5.4    | 1.0   | 0.9   | 1.2   | 1.5   | 1.5   | 1.9   | 1.9   | 1.5   | 1.4   | 1.9   | 0.9   | 1.1   |
|                                                                        | PrecStd     | 5.4    | 1.0   | 0.9   | 1.2   | 1.5   | 1.5   | 1.9   | 1.9   | 1.5   | 1.4   | 1.9   | 0.9   | 1.1   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 57.2  | 58.0  | 69.8  | 79.4  | 84.6  | 87.5  | 89.5  | 87.1  | 85.7  | 77.0  | 65.9  | 59.7  |
|                                                                        |             | MCWB   | 54.6  | 50.2  | 56.9  | 64.1  | 68.4  | 71.6  | 72.7  | 71.9  | 71.7  | 65.4  | 55.6  | 55.2  |
|                                                                        | 2%          | DB     | 49.7  | 49.0  | 59.2  | 72.2  | 80.6  | 83.8  | 85.7  | 83.4  | 81.0  | 71.3  | 60.4  | 50.8  |
|                                                                        |             | MCWB   | 46.2  | 42.6  | 50.6  | 58.4  | 66.6  | 68.1  | 71.0  | 70.4  | 69.5  | 62.4  | 52.8  | 48.0  |
|                                                                        | 5%          | DB     | 43.9  | 42.7  | 53.5  | 66.1  | 76.5  | 80.3  | 83.1  | 80.3  | 77.3  | 67.6  | 56.7  | 46.1  |
|                                                                        |             | MCWB   | 40.4  | 37.5  | 47.1  | 53.7  | 64.4  | 67.8  | 69.6  | 68.7  | 67.9  | 60.5  | 50.0  | 43.1  |
|                                                                        | 10%         | DB     | 38.9  | 37.6  | 47.9  | 61.1  | 72.1  | 76.8  | 80.2  | 77.6  | 73.1  | 63.4  | 52.4  | 41.8  |
|                                                                        |             | MCWB   | 36.3  | 33.4  | 42.4  | 52.1  | 61.2  | 65.3  | 68.5  | 67.4  | 65.6  | 57.9  | 46.9  | 39.1  |
|                                                                        | 0.4%        | WB     | 52.2  | 55.6  | 59.1  | 63.9  | 71.5  | 73.2  | 75.8  | 74.8  | 74.3  | 68.5  | 57.7  | 55.9  |
|                                                                        |             | MCDB   | 54.1  | 61.8  | 69.5  | 72.7  | 80.0  | 83.9  | 86.3  | 81.9  | 82.7  | 73.7  | 60.9  | 60.1  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 46.1  | 45.5  | 54.0  | 59.2  | 69.0  | 71.1  | 73.7  | 72.4  | 71.3  | 65.2  | 54.5  | 50.0  |
|                                                                        |             | MCDB   | 48.0  | 51.6  | 62.1  | 66.5  | 78.1  | 79.8  | 83.2  | 79.4  | 78.1  | 68.9  | 60.0  | 52.1  |
|                                                                        | 5%          | WB     | 40.8  | 39.4  | 48.9  | 55.4  | 66.4  | 69.2  | 72.1  | 70.6  | 69.4  | 62.5  | 50.8  | 45.2  |
|                                                                        |             | MCDB   | 42.9  | 43.9  | 53.7  | 62.1  | 73.9  | 76.6  | 80.5  | 77.0  | 76.9  | 66.9  | 55.5  | 47.1  |
|                                                                        | 10%         | WB     | 36.7  | 34.6  | 43.9  | 52.3  | 63.9  | 67.3  | 70.6  | 68.9  | 67.0  | 59.2  | 47.3  | 41.0  |
|                                                                        |             | MCDB   | 39.2  | 38.0  | 49.0  | 58.8  | 70.6  | 73.7  | 77.8  | 74.9  | 72.0  | 63.3  | 52.4  | 43.6  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 15.6  | 16.2  | 17.6  | 20.8  | 22.3  | 20.5  | 21.2  | 20.4  | 20.7  | 17.5  | 16.0  | 13.0  |
|                                                                        |             | MCDBR  | 18.7  | 21.8  | 25.2  | 31.4  | 27.8  | 25.4  | 24.6  | 22.9  | 24.0  | 23.0  | 22.6  | 17.0  |
|                                                                        | 5% WB       | MCWBR  | 17.3  | 16.8  | 17.0  | 17.8  | 14.4  | 13.7  | 11.8  | 11.9  | 12.6  | 14.4  | 15.5  | 14.2  |
|                                                                        |             | MCDWR  | 18.3  | 22.4  | 23.1  | 23.2  | 22.1  | 21.1  | 20.5  | 19.2  | 20.5  | 18.7  | 19.0  | 15.5  |
|                                                                        | 5% WB       | MCWBR  | 17.3  | 17.8  | 16.5  | 16.1  | 12.6  | 12.4  | 10.4  | 10.7  | 11.3  | 13.2  | 14.9  | 14.2  |
|                                                                        |             | MCDWR  | 17.3  | 17.8  | 16.5  | 16.1  | 12.6  | 12.4  | 10.4  | 10.7  | 11.3  | 13.2  | 14.9  | 14.2  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.291 | 0.309 | 0.330 | 0.357 | 0.384 | 0.434 | 0.441 | 0.423 | 0.362 | 0.347 | 0.305 | 0.285 |
|                                                                        | taud        |        | 2.375 | 2.328 | 2.389 | 2.373 | 2.336 | 2.213 | 2.207 | 2.244 | 2.401 | 2.447 | 2.517 | 2.472 |
|                                                                        | Ebn at Noon |        | 270   | 283   | 290   | 289   | 282   | 267   | 264   | 265   | 276   | 266   | 265   | 264   |
|                                                                        | Edh at Noon |        | 25    | 31    | 34    | 37    | 39    | 44    | 44    | 42    | 33    | 28    | 22    | 21    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 449   | 726   | 1069  | 1387  | 1661  | 1761  | 1810  | 1611  | 1290  | 808   | 524   | 357   |
|                                                                        | RadStd      |        | 39    | 103   | 96    | 162   | 165   | 164   | 140   | 93    | 131   | 90    | 66    | 48    |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   | (n)   |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (24 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | +167  | N/A   |

Nomenclature: See separate page